**KULLIYYAH OF INFORMATION & COMMUNICATION TECHNOLOGY**

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# CSCI 4343 Data Science
# SEMESTER I, 2025/2026
# SECTION 2

# Individual Assignment

Boycott Movement Analysis: Tracking Public Interest Through Google Search Data

| NAME | MATRIC NO. |
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| ALI ILHAN THANI BIN JALALUDIN | 2222253 |

**LECTURER**

DR. RAINI BINTI HASSAN

Research Question:

How has the boycott movement related to the Israeli-Palestinian conflict affected public search interest in Malaysia from January 2025 to October 2025?"

Real-World Problem:

The genocide of Palestinian by Israel have spark major consumer activism by many people especially Malaysian that lead us to boycott many products from Israel. Some corporations like McDonald's that are clearly supporting the economy of Zionist have faced significant backlash that lead to huge marginal losses in Muslim-majority countries like Malaysia.

This analysis helps us to determine the patterns of Malaysian public engagement that are boycotting products that support Israel. By tracking the certain keywords from google trends such as "boycott Israel", "BDS" (Boycott, Divestment and Sanctions) and "McDonald's boycott". We can monitor the sentiment of a certain brand and anticipate the impacts of boycotting toward the markets. This method of finding data from google trends will provide the objective and quantitative data on public interest that overcomes the limitations of traditional and online surveys that will have response bias based on where we conduct the traditional survey and where we share the link of the online survey.

Data Collection:

I collected real-world data from Google Trends API using R's gtrendsR package. I implemented two strategies to collect data. Firstly, by analyzing multiple boycott keywords within Malaysia such as "boycott Israel", "BDS" (Boycott, Divestment and Sanctions) and "McDonald's boycott". Also, I also compare the 'boycott Israel' searches across Malaysia, Indonesia, and Singapore. The timeline for this data collection is from 17/11/2024 to 17/11/2025."

```
# 1) DATA COLLECTION

# Strategy 1: Single country, multiple keywords
malaysia_data <- gtrends(
  keyword = c("boycott Israel", "BDS", "McDonald's boycott"),
  geo = "MY",  # Malaysia
  time = "today 12-m"
)

# Strategy 2: Single keyword, multiple countries
country_data <- gtrends(
  keyword = "boycott Israel",
  geo = c("MY", "ID", "SG"),  # Malaysia, Indonesia, Singapore
  time = "today 12-m"
)

print("REAL-WORLD DATA COLLECTED")
```

Data Processing:

Then, the raw data is processed and cleaned by removing missing values, converting dates of search keywords to proper format and making sure that the data types are consistent. Then, this dataset will be ready for the analysis. This is because we want to avoid potential errors and ensure our findings are reliable and trustworthy.

```
# 3) DATA PROCESSING
cat("Processing and cleaning data...\n")
clean_data <- malaysia_data$interest_over_time %>%
  filter(!is.na(hits)) %>%
  mutate(
    date = as.Date(date),
    keyword = as.factor(keyword)
  )
print("DATA PROCESSING COMPLETE")
```

Exploratory Data Analysis:

"Initial exploration revealed key patterns in search behavior. Summary statistics showed average interest levels and peak engagement periods for each boycott term, identifying when public attention was highest. For example, the summary statistics revealed that 'boycott Israel' maintained significantly higher average interest than brand-specific terms, indicating broader public concern with the general movement."

EDA exposed the key patterns in search behaviour among public interest. Summary statistics showed the average interest levels for boycott terms to identify when public attention was highest in this 10 months. For example, the summary statistics shows that the term “boycott Israel” always maintained significantly higher than other terms that we use, which are “BDA” and "McDonald's boycott". This can be seen that public concerns aim towards the general movement itself more than brand-specific terms.

Code:

```
# 3) EDA
cat("Performing exploratory analysis...\n")

# Summary statistics
summary_stats <- clean_data %>%
  group_by(keyword) %>%
  summarize(
    avg_interest = round(mean(hits, na.rm = TRUE), 2),
    max_interest = max(hits, na.rm = TRUE),
    peak_date = date[which.max(hits)],
    total_observations = n()
  )

print("SUMMARY STATISTICS")
print(summary_stats)
```

EDA Table:

```
[1] "SUMMARY STATISTICS"
> print(summary_stats)
# A tibble: 3 × 5
  keyword            avg_interest max_interest peak_date  total_observations
  <fct>                     <dbl>        <int> <date>                   <int>
1 BDS                        37.2           79 2025-10-05                  53
2 boycott Israel             16.8          100 2025-10-05                  53
3 McDonald's boycott          0.02           1 2025-11-02                  53
```

Machine Learning (Correlation and Trend analysis)

I applied correlation analysis to understand relationships between different boycott terms that are supported by trend analysis to identify and statistically validate significant patterns over time. This provided a data-driven narrative of how search interest in these terms both interacts and evolves."

For machine learning, correlation analysis is used to understand relationships between different boycott terms while trend analysis is used to validate significant patterns over time. This method provides a data interpretation on how search interest in these terms interacts and evolves with each other.

Code of correlation analysis:

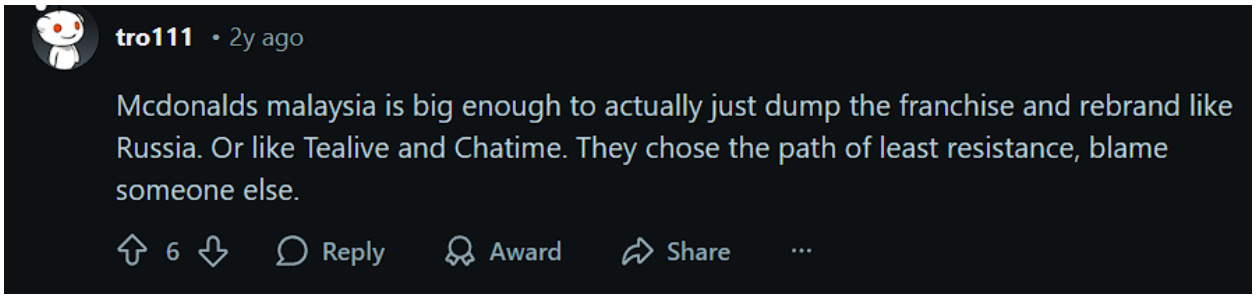
```
# 4) MACHINE LEARNING
print("Doing correlation and trend analysis\n")

# 1. Correlation analysis

all_keywords <- unique(clean_data$keyword)
cat("Keywords found:", paste(all_keywords, collapse = ", "), "\n\n")

# Create empty matrix
cor_matrix <- matrix(NA, nrow = length(all_keywords), ncol = length(all_keywords))
rownames(cor_matrix) <- all_keywords
colnames(cor_matrix) <- all_keywords

# Fill in the matrix from above
#for loop
for(i in 1:length(all_keywords)) {
  for(j in 1:length(all_keywords)) {
    # Get data for each keyword pair
    data_i <- clean_data$hits[clean_data$keyword == all_keywords[i]]
    data_j <- clean_data$hits[clean_data$keyword == all_keywords[j]]

    # Calculate correlation
    if(length(data_i) > 1 & length(data_j) > 1) {
      cor_matrix[i,j] <- round(cor(data_i, data_j, use = "complete.obs"), 3)
    }
  }
}

# Print the correlation table
print("Correlation Matrix:")
print(cor_matrix)
```

Correlation matrix table:

```
> print("Correlation Matrix:")
[1] "Correlation Matrix:"
> print(cor_matrix)
                   boycott Israel    BDS McDonald's boycott
boycott Israel              1.000  0.596              0.039
BDS                         0.596  1.000             -0.042
McDonald's boycott          0.039 -0.042              1.000
```

The result of the analysis shows a strong correlation which is 0.596 between the general term "boycott Israel" and the specific movement term “BDS”. The indication that can be seen from these google searches is that both of them are seen as closely related by the general public that are searching for the boycott’s terms.

Meanwhile, the contrast of the specific brand terms “McDonald’s Boycott” shows a weak correlation with both "boycott Israel" (r = 0.039) and "BDS" (r = -0.042). We can see that the general public searches did not see the boycott of specific corporate boycott movements as strong as the general boycott movement. It can be concluded that McDonald’s boycott appears to be less important compared to the general and specific movement term in the whole scheme of boycotting Israel products boycott campaign. This can be seen from some discourse online between the general public in Malaysia that argues that McDonald Malaysia is Muslim owned so it doesn't need to be boycotted. In contrast, there are also sides that ask for the boycott of mcdonald because of the royalty that needs to be paid for the Israel company.



Code of Trend Analysis:

```
# Create empty data frame
trend_results <- data.frame(
  keyword = character(),
  slope = numeric(),
  p_value = numeric(),
  trend = character(),
  stringsAsFactors = FALSE
)

#for loop
for(term in all_keywords) {
  term_data <- clean_data[clean_data$keyword == term, ]

  if(nrow(term_data) > 1) {
    # Simple linear model
    model <- lm(hits ~ as.numeric(date), data = term_data)

    # Get results
    slope_val <- round(coef(model)[2], 4)
    p_val <- round(summary(model)$coefficients[2,4], 4)
    trend_dir <- ifelse(slope_val > 0, "INCREASING", "DECREASING")

    # Add to results
    trend_results <- rbind(trend_results,
                           data.frame(keyword = term,
                                      slope = slope_val,
                                      p_value = p_val,
                                      trend = trend_dir))
  }
}

# Printing trend analysis
print("Trend Analysis:")
print(trend_results)
cat("\n")
```

Table of trend analysis:

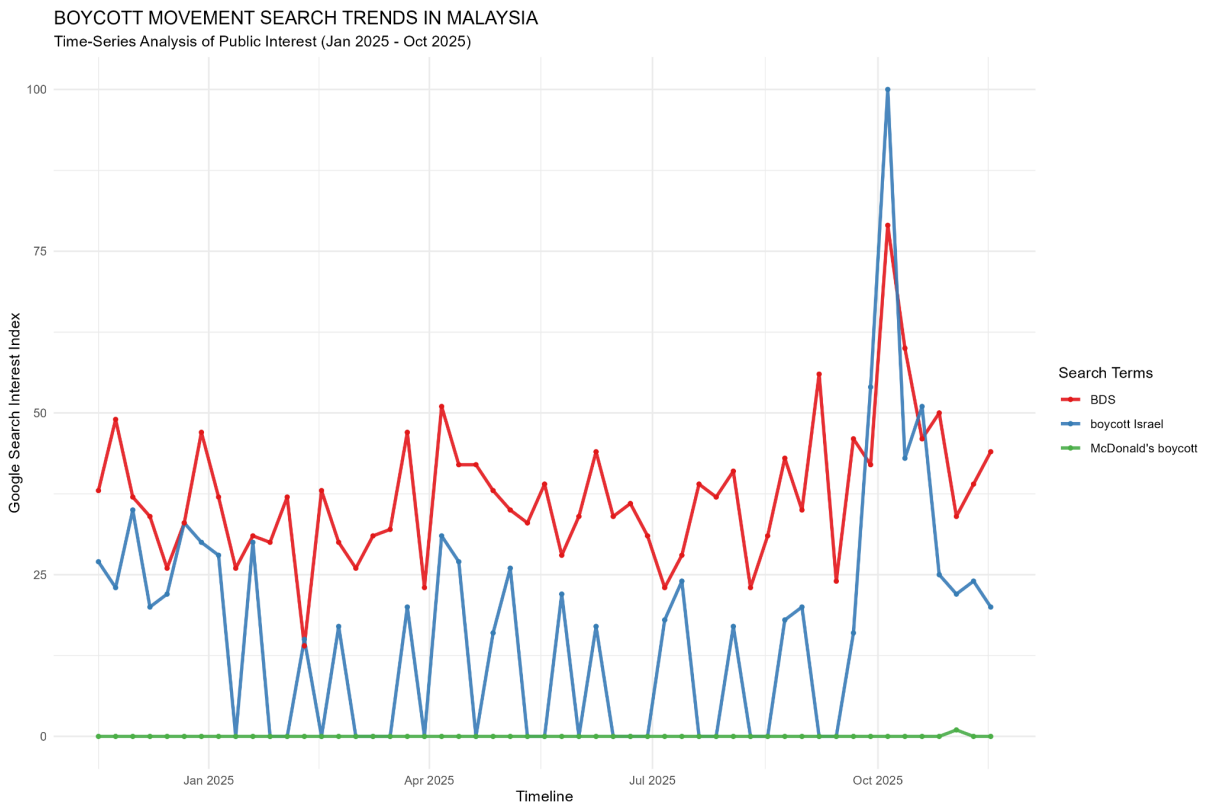
```
[1] "Trend Analysis:"
> print(trend_results)
                              keyword  slope p_value      trend
as.numeric(date)       boycott Israel 0.0265  0.2708 INCREASING
as.numeric(date)1                 BDS 0.0319  0.0183 INCREASING
as.numeric(date)2 McDonald's boycott 0.0003  0.1176 INCREASING
> cat("\n")
```

The function of trend analysis is to share the popularity of the keyword compared to correlation analysis that used to correlate keywords with each other. Even though the results of trend analysis are almost the same as correlation analysis. The slope of the trend is steadily increasing for both “boycott Israel” and “BDS” meanwhile there's almost no change in the term “McDonald’s Boycott” with 0.003 of slope even though it is increasing. We can conclude that the term “McDonald’s Boycott” is not as famous as the other two terms in the eyes of the public.
Graphic visualization:

To understand the statistical analysis, all the search trends are visualized to make it easier to understand. The graphic visualization that were used in this project is divided by four parts, which are:

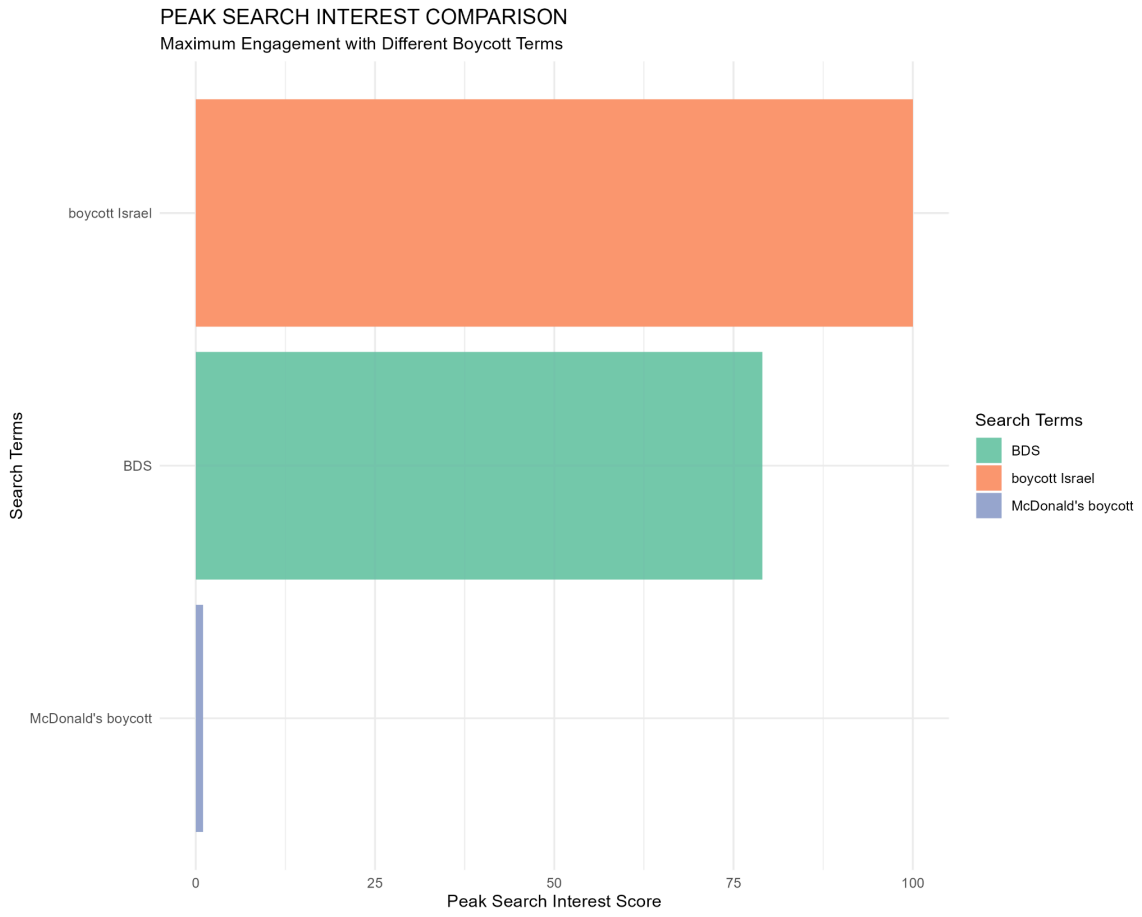
- Time series analysis
- Peak analysis
- Monthly analysis
- Country comparison

Time series analysis:



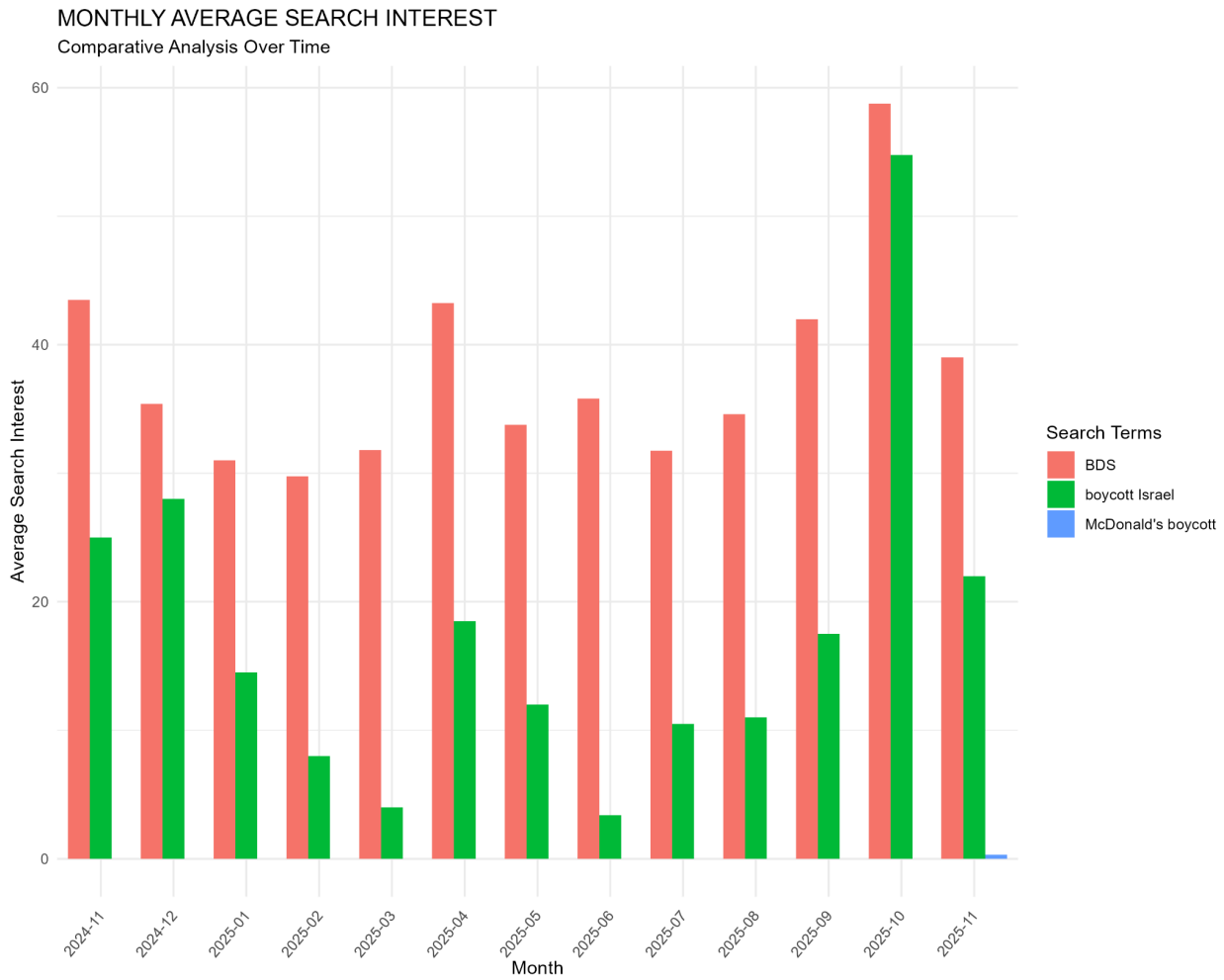

The time series analysis reveals that all three keywords show an increasing and decreasing trend except for “Mcdonald’s boycott” where it stays almost the same except for late October of this year where it goes up a little bit. For “BDS” we can see that the trends are always above 0, this means that the “BDS” statistic is always significant. This event shows that “BDS” is a robust term that is reliable for the public's attention. Meanwhile, “boycott Israel” is not as significant as “BDS” but more than “McDonald’s boycott” but at its peak, it's the highest google search index compared to others.

Peak analysis:



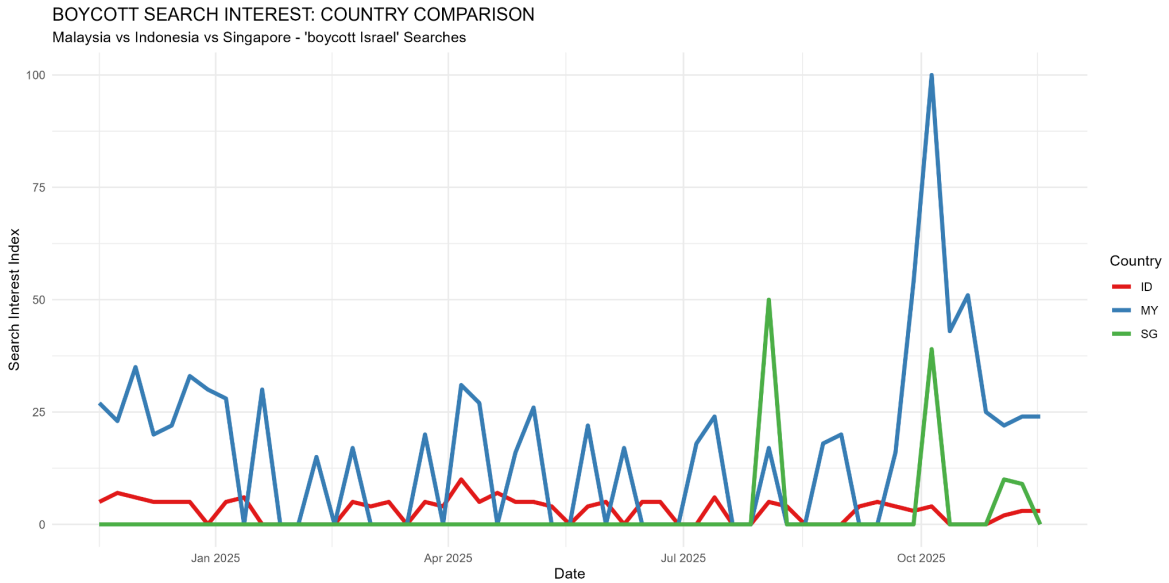

The peak analysis graph shows that the general term “boycott Israel” generates the highest level of engagement in a certain time period which is around October 2025, this can be backed up from the time series analysis graph. It also generates the maximum score for google search interest index. Then, it is followed by “BDS” which are more lenient and stable compared to other terms. That is why the “BDS” only managed to get around a 75 index score. Lastly we have the “McDonald’s boycott” term where its peak index did not even manage to go more than 5.

Monthly Analysis:




This graph is the same as the time series analysis but shows it in a bar chart compared to the lines. This graph can make it easier to read the time analysis because it is a more systematic method that shows each term one by one without lines overlapping each other.

Country Comparison:




This is an additional graphic visualization that indicates the term “boycott Israel” in three different countries which are Malaysia, Indonesia and Singapore. This graph is made to see which country has the highest average search index and the highest peak index. It can be seen that Malaysia topples both of the index that we are searching for. Primarily, the Malaysian government has had a long-standing support towards Palestine and maintained a critical stance towards Zionism ever since the era of our former Prime Minister Mahathir Mohammad(2016) until our current one, Anwar Ibrahim(2023)

Meanwhile, for Indonesia the government is not as proactive as Malaysia in support towards Palestine and critical stance towards Zionism because of the Indonesian government’s corruption(Tambunan, 2022) unlike the general public of Indonesian that supports Palestine by doing peaceful protest for the freedom of Palestine  The Straits Times. (2023, Nov 05). Lastly, we can see that Singapore's search index is very low, with only 3 periods of time where it was not 0. This can be seen from Singapore’s goverment  policy of maintaining neutrality among other governments to balance the diplomatic stance. The Singaporean government has called out for peaceful resolution and provides humanitarian aid, but avoid condemnation against Israel to maintain their relationship with both Western allies and its neighbouring neighbours.

In conclusion, the contrast between these 3 different countries in southeast asia  are  not just a reflection of the general public towards the boycott. It is also a representation of the government's political environments towards the Israel-Palestine Genocide. The data visualizes a clear difference between this issue. Starting with Malaysia, our country is a government that highly supports Palestine and so do the people. Meanwhile, Indonesia’s support mainly comes from the people, not from the government. Lastly, Singapore’s government maintains an almost neutral stance in this genocide just by supporting Palestine without condemning Israel.

Conclusion:

The data science project has managed to trace and analyze the boycott movement of the Israel products from late 2024 to late 2025. This is done by using the data from Google Trends and applying some machine learning techniques such as EDA(summary statistics), correlation analysis and trend analysis. Some graphs were also made to make the analysis easier to read. It is also noted that the result may vary when running the code because the timeline of data collection is from the exact day one year ago until the day the code is run.

The analysis demonstrates that the public's search is mainly focused on the general political movement. The strong correlation between "boycott Israel" and "BDS" reveals a significant statistical trend for the general and organized terms. In contrast, a specific movement such as "McDonald's Boycott" showed a weak correlation between the general and organized term. This indicates that the boycott movement is likely an event-specific discourse rather than a sustainable interest among the public.

The data analysis also shows the correlation between 3 countries in supporting the boycott. Showcasing the significance of boycotts in each country in a different way. The boycott index shows that Malaysia is first, followed by Indonesia and Singapore.

References:

Tambunan, D. (2022). Indonesia under threat: The danger of corruption to political legitimacy. Asian Journal of Comparative Politics, 8(1), 112–140. https://doi.org/10.1177/20578911221124965

Anwar Ibrahim. (2023, October 24). Anwar Ibrahim's support towards Palestine. [Video]. https://youtu.be/V1ONCQa5XZE?si=VYU91JSTIluVo6eW

Mahathir Mohamad. (2016, June 7). *Mahathir Mohammad critical stance towards the zionist [Video]. YouTube. https://youtu.be/RnYUGuhtNk4?si=7PWav2GWC-4jnUrx

The Straits Times. (2023, Nov 05). Indonesians join peaceful rallies in solidarity with Palestine. The Straits Times. https://www.straitstimes.com/asia/se-asia/indonesians-join-peaceful-rally-in-solidarity-with-palestine